

Metro Ridership Forecasting using Inter-Station-Aware Transformer Networks

Big Data Course Presentation (10 minutes max)

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Big Data Course



Presentation Roadmap

- ▶ Problem and Motivation
- ▶ ISA-TF Architecture
- ▶ Experimental Evaluation
- ▶ Key Insights and Discussion



[1/12] Problem Context (Presenter 1)

- ▶ Metro systems need station-level inflow/outflow forecasts for operations.
- ▶ Accurate forecasts improve:
 - train scheduling
 - staffing
 - maintenance planning
- ▶ Poor forecasts → congestion and delays
- ▶ The task is hard because demand is both **temporal** and **network-dependent**.



[2/12] Research Gap and Paper Idea (Presenter 1)

- ▶ Prior methods include statistical, machine learning, and hybrid approaches.
- ▶ Many RNN/GCN-based methods are strong for short horizon but can degrade for longer horizon.
- ▶ This paper proposes **ISA-TF**: a unified transformer framework for metro ridership forecasting.
- ▶ Core idea: fuse ridership history with inter-station topology information.



[3/12] Claimed Contributions (Presenter 1)

- ▶ New **Inter-Station-Aware Transformer Networks** architecture.
- ▶ Uses three metro graph views: physical, semantic similarity, and correlation.
- ▶ Evaluated on SHMetro and HZMetro benchmarks.
- ▶ Reports stronger long-horizon performance and a much lower parameter count.



[4/12] Problem Formulation (Presenter 2)

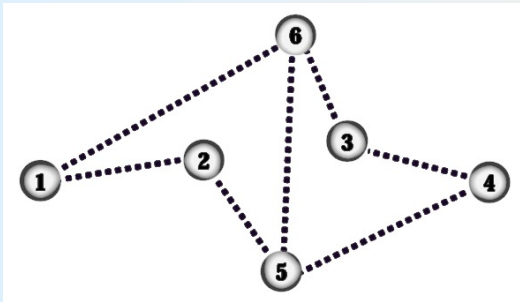
- ▶ Input: past ridership sequence and metro graph:

$$\{D_{t-n+1}^N, \dots, D_t^N\}, G \rightarrow \{\hat{D}_{t+1}^N, \dots, \hat{D}_{t+m}^N\}$$

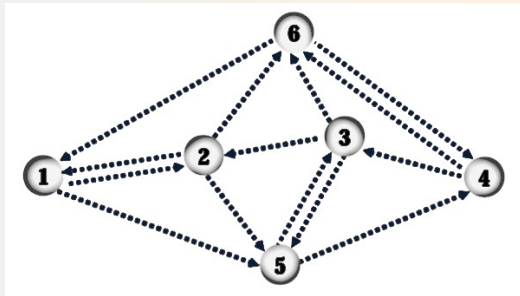
- ▶ Each station signal contains two values: inflow and outflow.
- ▶ In experiments: $n = 4$ past intervals (60 min total), $m = 4$ future intervals.
- ▶ Inference is autoregressive, so horizons longer than 60 min are possible.



[5/12] Metro Graph Representation (Presenter 2)



Physical topology view



Enhanced inter-station relation view

Explanation: The paper represents the metro network with complementary graph structures, not only direct neighboring links. This helps the model capture both local structure and richer cross-station interaction patterns, which improves forecasting robustness when passenger flow dependencies are not purely geographic.



Why Use Transformers?

- ▶ RNN-based models struggle with long-range temporal dependencies
- ▶ Transformers use **self-attention** to capture long-term patterns
- ▶ Parallel computation improves scalability
- ▶ ISA-TF integrates network topology into the transformer framework



[6/12] ISA-TF Architecture (Presenter 2)

- ▶ Transformer encoder–decoder backbone
- ▶ Encoder processes historical ridership
- ▶ Graph information fused using:
 - M_p physical topology
 - M_s semantic similarity
 - M_c correlation matrix
- ▶ Decoder predicts future ridership autoregressively



ISA-TF Architecture Pipeline



[7/12] Datasets (Presenter 3)

Property	SHMetro	HZMetro
Period	Jul-Sep 2016	Jan 1-25, 2019
Stations	288	80
Physical edges	958	248
Avg daily passengers	8.82M	2.35M
Interval	15 minutes	15 minutes

SHMetro has 811.8M recorded transactions.



[8/12] Experimental Setup (Presenter 3)

- ▶ Preprocessing creates the three graph types per dataset.
- ▶ Training details: Adam optimizer, 250 epochs, L_2 loss.
- ▶ Model hyperparameters: embedding size 512, 8 attention heads.
- ▶ Evaluation metrics: RMSE(Root Mean Square Error) and MAE(Mean Absolute Error).
- ▶ Baselines: HM(Historical Mean),
GB(Gradient Boosting),
MLP(Multiple Layer Perceptron),
RNN-GRU(recurrent neural network-Gated Recurrent Unit, it consists of two GRU layers with its hidden units set to 256.),
STG2Seq(Spatial-temporal Graph to Sequence),
DCRNN(Diffusion Convolutional Recurrent Neural Network),
PVCGRN(Physical-Virtual Collaboration Graph Network).



[9/12] Main Results (Presenter 3)

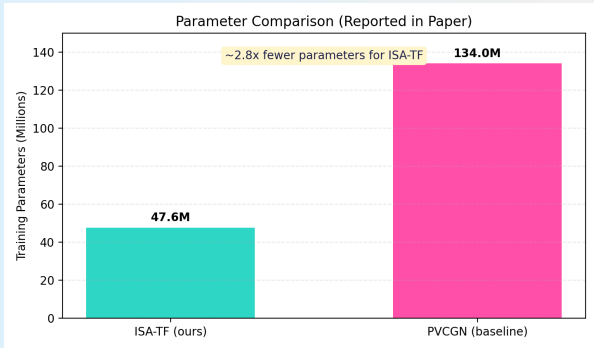
Dataset	Horizon	ISA-TF (RMSE/MAE)	PVCGN (RMSE/MAE)
SHMetro	60 min	48.43/23.87	55.27/26.29
HZMetro	60 min	41.68/24.91	42.61/24.93
SHMetro	15 min	45.85/ 23.05	44.97 /23.29

- ▶ Strong long-horizon robustness is the key result.
- ▶ Short horizon remains competitive with the prior SOTA.

Key Result: ISA-TF performs best on long-horizon prediction (45–60 min).



[10/12] Efficiency Comparison (Presenter 4)



Explanation: Based on the paper's reported values, ISA-TF (Inter-Station-Aware Transformer Networks model) uses 47.6M parameters while PVCGN (Physical-Virtual Collaboration Graph Network) requires about 134M. This means ISA-TF is around 2.8x lighter, which is important for faster training, lower memory use, and easier deployment while keeping competitive or better forecasting performance.

Source: values reported by the paper (Figure 2 discussion and results section).



- ▶ **Strengths:** unified architecture, good long-horizon behavior, parameter efficiency.
- ▶ **Caveat:** evaluation is on two datasets only; generalization should be tested further.
- ▶ **Possible extensions:**
 - Include weather/events/holidays as exogenous features.
 - Add uncertainty intervals, not only point forecasts.
 - Evaluate transfer across cities with different network shapes.



[12/12] Conclusion and Q&A (Presenter 4)

- ▶ ISA-TF combines ridership history with topology-aware graph information.
- ▶ It matches or beats prior methods, especially for 45–60 minute horizons.
- ▶ It achieves this with about 3x fewer parameters than PVCGRN.

Takeaway: topology-aware transformers are a strong option for scalable metro ridership forecasting.

Reference: K. Saleh, A.-S. Mihaita, and Y. Ou, 2023 IEEE ITSC (Manuscript 470 preprint).



Thank You

Questions?



1.1 What is Time-Series Data?

Time-Series is data collected over time, in chronological order.

The paper frames ridership prediction as a time-series forecasting problem. Imagine counting people at a door every 15 minutes.

- ▶ t represents the *current* time interval (e.g., 10:00 AM).
- ▶ $t - 1$ is the *previous* interval (9:45 AM).
- ▶ $t + 1$ is the *next* interval (10:15 AM).

When the paper says it uses $n = 4$ historical steps to predict $m = 4$ future steps, it means: "*I will look at the last 60 minutes (four 15-min chunks) to guess the next 60 minutes.*"



1.2 The Problem: Time-Series Forecasting

What are we trying to do? Predict how many passengers will enter and exit a metro station in the future based on past data.

The Notation (The Math):

- ▶ Past data sequence (length n): $(D_{t-n+1}^N, D_{t-n+2}^N, \dots, D_t^N)$.
- ▶ Future data sequence (length m): $(\hat{D}_{t+1}^N, \hat{D}_{t+2}^N, \dots, \hat{D}_{t+m}^N)$.
- ▶ N is the total number of stations.
- ▶ $D \in \mathbb{R}^2$ means the data has two values:
 - **Inflow Count:** People entering the station to ride.
 - **Outflow Count:** People leaving the station into the city.



2.1 What is a Neural Network?

Goal: Learn a function that maps inputs to outputs.

Example in this paper:

Past ridership data $\rightarrow$ Future ridership prediction

A neural network is a stack of mathematical functions $y = f(x)$, where:

- ▶ x = input data
- ▶ f = learned function
- ▶ y = predicted output

Instead of manually designing the function, the network **learns it from data**.



2.2 Weights and Parameters

Neural networks learn by adjusting internal numbers called **parameters** or **weights**.

Example of a simple neuron:

$$y = w_1x_1 + w_2x_2 + b$$

Where:

- ▶ x_1, x_2 = input features
- ▶ w_1, w_2 = weights (parameters to learn)
- ▶ b = bias

During training, the model changes these weights so predictions become more accurate.



2.3 Forward Pass: Making a Prediction

Training a neural network starts with a **forward pass**.

Steps:

1. Input data enters the network.
2. Each layer applies a mathematical transformation.
3. The network produces a prediction.

Example:

Past metro data $\rightarrow$ Neural network $\rightarrow \hat{y}$ (predicted ridership)

At this stage, the prediction is usually wrong because the weights are not yet trained.



2.4 Loss Function: Measuring Error

After making a prediction, we measure how wrong it is using a **loss function**.

Example used in the paper: L2 loss

$$L = (\hat{y} - y)^2$$

Where:

- ▶ y = true historical value
- ▶ $\hat{y}$ = AI's predicted value

Goal of training: **Minimize the loss**. The smaller the loss, the better the prediction.



3.1 What is Backpropagation?

The Concept: If the "Forward Pass" is guessing, **Backpropagation** is learning from the mistake. It sends the error backwards to update the weights.

1. FORWARD (Guessing):

Data ---> [Weight A] ---> [Weight B] ---> Guess (100 people)
Real Answer (150)
ERROR = 50

2. BACKWARD (Learning via Backpropagation):

Data <--- [Update A] <--- [Update B] <--- ERROR = 50
(Blame A) (Blame B)

- ▶ It uses the **Chain Rule** from calculus to calculate "Gradients".
- ▶ It asks: *"How much did Weight B contribute to the error? How about A?"*
- ▶ The Optimizer uses this "blame" to tweak the weights.



3.2 Gradients and Backpropagation (The Math)

To improve predictions, the network adjusts its weights using **gradients**.

A gradient tells us:

"If we slightly change this weight, how will the loss change?"

Mathematically:

$$\frac{\partial L}{\partial w}$$

The exact Backpropagation sequence: 1. Compute prediction (Forward Pass) → 2. Measure loss → 3. Compute gradients (Backward Pass) → 4. Update weights.
This repeats thousands of times.



3.3 How AI Learns: The "Mountain" Analogy

Imagine the AI is standing at the top of a mountain, blindfolded.

- ▶ **Top of the mountain:** 100% wrong (high error/loss).
- ▶ **Bottom of the valley:** 100% right (zero error).

The AI wants to walk down to the bottom.

- ▶ The **Loss Function** tells the AI how high up the mountain it currently is.
- ▶ The **Optimizer** is the strategy the AI uses to feel the ground with its foot and decide which direction to step next to go downhill.



3.4 What is the Adam Optimizer?

An **Optimizer** is the mathematical engine that updates the weights based on the gradients.

Adam (Adaptive Moment Estimation) is a popular, smart optimizer.

- ▶ Instead of taking the same size step every time, Adam adapts.
- ▶ If the mountain is steep, Adam takes big, fast steps.
- ▶ If the AI feels it is getting close to the bottom (the right answer), Adam automatically takes tiny, careful steps so it doesn't overshoot the target.



3.5 Epochs and The Training Loop

What is an Epoch? Think of the historical metro dataset as a giant stack of flashcards. An **Epoch** is exactly ONE full pass through the entire stack. This model trained for **250 epochs** (it reviewed the history 250 times).

The Full Training Loop:

1. **Guess:** AI looks at 60 mins of past data, guesses next 60 mins.
2. **Check Error:** Compares guess to the real answer (L2-loss).
3. **Learn:** Backpropagation finds gradients; Adam tweaks weights.
4. **Repeat:** Do this for all data (1 Epoch). Repeat 250 times!



4.1 How Gradients Flow & Why They Vanish

During training, the error signal travels backward. Each layer multiplies the gradient by a fraction. If many layers are stacked, the gradient shrinks rapidly.

$$\text{Layer 4 gradient} = 1.0 \times 0.5 = 0.5$$

$$\text{Layer 3 gradient} = 0.5 \times 0.5 = 0.25$$

$$\text{Layer 2 gradient} = 0.25 \times 0.5 = 0.125$$

$$\text{Layer 1 gradient} = 0.125 \times 0.5 = 0.0625 \quad \leftarrow \text{Almost Zero!}$$

The Vanishing Gradient Problem: When gradients become extremely small near the input layers:

- ▶ Weight updates become tiny.
- ▶ Early layers stop learning entirely.
- ▶ Training becomes very slow or stalls.



4.2 Activation Functions in Neural Networks

Motivation: Neural networks naturally apply simple linear transformations (straight lines). If only linear operations were used, the network could never learn complex patterns (like sudden crowd spikes at a metro).

Solution: Activation Functions

- ▶ Activation functions introduce **nonlinearity**.
- ▶ They act as "gates" deciding if a neuron should fire or not.
- ▶ Without them, deep networks are useless; with them, networks can approximate almost any complex real-world data curve.



4.3 Sigmoid vs. ReLU

Older models used the **Sigmoid** function (an S-curve between 0 and 1). At the flat edges of the curve, the slope is nearly 0. This caused massive Vanishing Gradients.

The Fix: ReLU (Rectified Linear Unit)

$$\text{ReLU}(x) = \max(0, x)$$

- ▶ If $x \leq 0$, output is 0. If $x > 0$, output is x .
- ▶ Because the slope for positive numbers is a constant straight line, gradients flow perfectly without shrinking.

ReLU | / Gradient is constant for positive inputs.
 | / This stops gradients from vanishing!
 |_----/
 +-----



5.1 Graph Theory: Mapping the Metro

In Big Data, we represent networks using **Graphs**. This paper uses THREE different graphs to give the AI the full picture:

1. Physical Graph
(Physical Tracks)

```
A --- B
|       |
C --- D
```

(Who is next door?)

2. Semantic Graph
(Similar Behaviors)

```
A       B
|       |
C       D
```

(Who is busy at the
same time?)

3. Correlation Graph
(Actual Travel Routes)

```
A ===== B
 \         /
  C       D
```

(Where do people
actually travel?)

An **Edge Weight** is a number showing how strongly two stations are connected in these specific views.



5.2 Physical (M_p) & Semantic (M_s) Graphs

Physical Edge Weight (M_p): Are these stations directly connected by a train track?

- ▶ Base score is 1 if connected, 0 if not. Formula standardizes it to fractions.
- ▶ *Why it matters:* Captures direct crowd spillover to the next station.

Semantic Similarity (M_s) & DTW: Do these stations have the same "vibe"? (e.g., two business districts far apart that both peak at 5:00 PM).

- ▶ Uses **Dynamic Time Warping (DTW)**: An algorithm that matches similar time-series curves even if one is slightly delayed.
- ▶ *Why it matters:* Connects stations based on human behavior, not geography.



5.3 Correlation Matrix (M_c)

Correlation Matrix (M_c): Are people actually riding the train directly from Station A to Station B?

- ▶ The physical graph shows the tracks.
- ▶ The semantic graph shows the schedule/behavior.
- ▶ **The correlation graph shows the actual humans.** It counts total travels starting at A and ending at B.

The Math: Takes trips between A and B (R_c) and divides by total trips in network.

$$M_c(a, b) = \frac{R_c(a, b)}{\sum_{k=1}^N R_c(a, k)}$$

Why it matters: Directly maps where crowds are flowing in real-time.



6.1 RNNs vs. Transformers

The Old Way: RNNs (Recurrent Neural Networks)

- ▶ RNNs read data chronologically, step-by-step.
- ▶ *The Flaw:* They struggle to remember old information. They suffer from vanishing gradients over long time horizons.

The New Way: Transformers

- ▶ Transformers don't read step-by-step. They look at the entire sequence at once.
- ▶ They use **Self-Attention** to score which past time steps are the most important for predicting the future.



6.2 Embeddings: Converting Data into Vectors

Problem: Neural networks cannot process raw concepts, words, or raw time steps easily.

Solution: **Embeddings** transform each time-step token into a dense vector of real numbers.

- ▶ Similar data patterns obtain similar vectors in the mathematical space.
- ▶ These embeddings become the starting input used to compute **Query (Q)**, **Key (K)**, and **Value (V)** vectors in the Transformer.



6.3 Unpacking "Self-Attention"

Analogy: Reading a sentence. If you read: "The **bank** of the **river** was muddy." To understand the word "bank", your brain pays *attention* to the word "river" (not a money bank).

Step 1: Q, K, V Vectors From the embedding, the network creates 3 vectors:

- ▶ **Query (Q)** – what the token is looking for.
- ▶ **Key (K)** – what the token represents.
- ▶ **Value (V)** – the actual information the token carries.

Step 2: Attention Scores The model computes similarity by multiplying Query and Key ($Q \cdot K$). Higher scores mean the model should pay more attention to that time step.



6.4 What is Softmax?

The Concept: Softmax is a mathematical function that takes raw, random Attention scores and converts them into **probabilities** that sum to 1.0 (100%).

Raw Attention Scores		Softmax Math		Final Probabilities	
Time 4:30 PM:	3.2	--->	$e^{(3.2)} / \text{Sum}$	--->	84% (Focus here!)
Time 4:45 PM:	1.0	--->	$e^{(1.0)} / \text{Sum}$	--->	11% (A little)
Time 4:00 PM:	-0.5	--->	$e^{(-0.5)} / \text{Sum}$	--->	5% (Ignore)
TOTAL:			3.7		100%

Step 3: Final Output The Softmax probabilities are multiplied by the **Value (V)** vectors. The AI successfully filters out the noise and focuses 84% of its computing power on the 4:30 PM data spike!



6.5 Self-Attention Mechanism (The Math)

Computation Summary:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- ▶ Q = Query matrix, K = Key matrix, V = Value matrix.
- ▶ The dot product QK^T measures similarity between time steps.
- ▶ $\sqrt{d_k}$ scales it down so gradients don't explode.
- ▶ Softmax converts to weights (summing to 1).
- ▶ Result is multiplied by V to get the final representation.



7.1 The Paper's Invention: ISA-TF

The **Inter-Station-Aware Transformer Networks (ISA-TF)** combines the time-reading power of Transformers with the spatial map of the 3 Graphs.

[PAST DATA]	-->	(ENCODER)	-->	[FUSION]	-->	(DECODER)	-->	[FUTURE DATA]
Last 60 mins		Finds Time Patterns		Mixes Time with 3 Graphs		Guesses the Future		Next 60 mins

- ▶ **Encoder:** Uses self-attention to capture time patterns.
- ▶ **Fusion:** Fuses encoder output with the 3 metro graphs.
- ▶ **Decoder:** Predicts the future auto-regressively.



7.2 Deep Dive: Encoder & Feed-Forward

Encoder:

- ▶ Adds "Positional Encoding" so the Transformer knows the order of time (since it doesn't read left-to-right like an RNN).
- ▶ Uses Self-Attention to find patterns.
- ▶ Passes data through a **Feed-Forward Neural Network (FFN)**:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Notice the $\max(0, \dots)$? That is the ReLU activation function preventing vanishing gradients!



7.3 Deep Dive: Fusion & Decoder

Intermediate Fusion (Concatenation):

- ▶ It simply glues vectors together.
- ▶ Example: Time Vector $[0.5, 0.2]$ + Graph Vector $[0.7, 0.1]$ = Fused Vector $[0.5, 0.2, 0.7, 0.1]$.
- ▶ Now the AI understands both TIME and SPACE simultaneously.

Decoder:

- ▶ Uses **Masked** Self-Attention (so it can't "cheat" and look at future answers during training).
- ▶ Uses Cross-Attention to look back at the Fused Vector.
- ▶ Predicts the next 15 minutes, feeds that prediction back into itself, predicts the next 15 minutes, etc. (Auto-regressive).



8.1 Evaluation Metrics (RMSE & MAE)

RMSE (Root Mean Square Error): Heavily punishes large errors because differences are squared before averaging. Good for preventing massive congestion mistakes.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{D}_i - D_i)^2}$$

MAE (Mean Absolute Error): The simple average of how far off the AI's guesses were.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{D}_i - D_i|$$

Note: $\hat{D}_i$ is predicted ridership, D_i is actual ground-truth.



8.2 Decoding the "Baselines" (Competitors)

The paper compares ISA-TF against older models:

- ▶ **HM (Historical Mean):** Just averages past data.
- ▶ **GB (Gradient Boosting):** A classic ML method combining decision trees.
- ▶ **RNN-GRU:** A recurrent network that processes step-by-step (forgets long-term trends).
- ▶ **DCRNN / PVCGRN:** Advanced models that rely on *Convolutions* and graph networks.



8.3 Why do "Parameters" Matter?

The ISA-TF model uses ~ 47.6 Million parameters. The previous best model (PVCGRN) required ~ 134 Million.

- ▶ Think of parameters as the "gears" inside the AI's engine.
- ▶ **Why fewer is better:** A model with fewer parameters trains faster, uses less RAM/electricity, and is much easier for a city to deploy in the real world while getting the exact same or better accuracy!



8.4 Baseline Concept: What is Convolution?

Many baseline models (like DCRNN) use Convolutions instead of Attention.

Definition: Convolution slides a small mathematical filter (kernel) over data to extract local patterns.

- ▶ Example Input: $[2, 4, 6, 8]$
- ▶ Filter: $[1, 0]$
- ▶ It scans: $(2 \cdot 1 + 4 \cdot 0) = 2 \rightarrow (4 \cdot 1 + 6 \cdot 0) = 4 \dots$

While Convolutions are great for finding local spikes (like an edge in an image), they struggle to connect Station A to Station Z simultaneously, which is why the Transformer's Global Attention usually wins for long-range tasks.



9.1 Equation-Level Mapping for Slide [5/12]

The paper defines three normalized edge-weight matrices (Eq. 1–3):

- ▶ Physical topology graph

$$M_p(a, b) = \frac{E(a, b)}{\sum_{k=1}^N E(a, k)}$$

- ▶ Semantic similarity graph (DTW-based)

$$M_s(a, b) = \frac{F_s(a, b)}{\sum_{k=1}^N F_s(a, k)}$$

- ▶ Correlation graph (Origin–Destination travel interactions)

$$M_c(a, b) = \frac{R_c(a, b)}{\sum_{k=1}^N R_c(a, k)}$$

Interpretation: ISA-TF captures both nearby-station effects and cross-station flow interactions beyond direct physical links.



9.2 Deep Dive: Physical Topology Graph (M_p)

The Concept: Geographical Connections

- ▶ This graph represents the actual, physical train tracks.
- ▶ If Station A and Station B are next to each other on the same line, they get a base score of 1 ($E(a, b) = 1$).
- ▶ If they are not directly connected, they get a 0.

Why does the AI need this?

- ▶ **Spillover Effect:** Passenger flow naturally moves down the line. If Station A gets extremely crowded at 8:00 AM, the AI uses this graph to predict that the next physical station down the track will likely see a surge a few minutes later.



9.3 Deep Dive: Semantic Similarity Graph (M_s)

The Concept: Behavioral Connections

- ▶ This graph ignores geography and looks at **station function**.
- ▶ Example: Two business-district stations on opposite sides of the city might both experience massive outflow at 9:00 AM and massive inflow at 5:00 PM.

How it's calculated: Dynamic Time Warping (DTW)

- ▶ DTW is an algorithm that compares the shape of two time-series curves.
- ▶ Even if Station A peaks at 5:00 PM and Station B peaks at 5:15 PM, DTW recognizes they share the same "vibe" or pattern and links them mathematically.
- ▶ **Why it matters:** It helps the AI learn city-wide demographic habits.



9.4 Deep Dive: Correlation Graph (M_c)

The Concept: Actual Human Movement

- ▶ This graph tracks the **Origin-Destination (OD)** pairs.
- ▶ It simply counts how many historical passengers tapped their ticket to enter at Station A, and tapped out to exit at Station B.

Why does the AI need this?

- ▶ The Physical Graph shows tracks, and the Semantic Graph shows schedules, but the Correlation Graph shows **actual travel paths**.
- ▶ If historical data shows that 40% of people who leave a stadium at Station A travel directly to a transfer hub at Station B, this graph gives the AI the exact mathematical percentage it needs to predict the crowd at Station B.